

SAMUEL C MOGEN

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Institute of Arctic and Alpine Research, University of Colorado

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EDUCATION

University of Colorado Boulder	<i>Boulder, CO</i>
Ph.D. Candidate in Atmospheric & Oceanic Sciences	<i>May 2025</i>
Dissertation title: <i>Dynamics and Predictability of Marine Ecosystem Stressors</i>	
Dissertation advisor: <i>Nicole Lovenduski</i>	
M.S. in Atmospheric & Oceanic Sciences	<i>May 2022</i>
University of Virginia	<i>Charlottesville, VA</i>
B.S. in Environmental Science	<i>2020</i>
<i>with Highest Distinction (Summa Cum Laude)</i>	
B.A. in Global Environments and Sustainability	<i>2020</i>

WORK EXPERIENCE

University of Colorado Boulder	<i>Boulder, CO</i>
Post-Doctoral Associate, Institute of Arctic and Alpine Research	<i>2025-present</i>
University of Colorado Boulder	<i>Boulder, CO</i>
Graduate Research Assistant, Institute of Arctic and Alpine Research	<i>2021-2025</i>
University of Virginia	<i>Charlottesville, VA</i>
Undergraduate Research Assistant, Computational Biogeochemistry Lab	<i>2019-2020</i>
UVA Nitrogen Working Group	<i>Charlottesville, VA</i>
Co-Chair and Researcher	<i>2017-2020</i>
NOAA Pacific Marine Environmental Lab	<i>Seattle, WA</i>
NOAA Hollings Scholar, Physical Sciences Division	<i>Summer 2019</i>

PUBLICATIONS

In preparation:

7. **Mogen, S.C.**, Lovenduski, N.S., Krumhardt, K., Yeager, N.S.. *in preparation*. Dynamics and predictability of marine phytoplankton extremes. *Geophysical Research Letters*.
6. Kohlmann, K., **Mogen, S.C.**, Cohen, J.T., Lovenduski, N.S., Sharp, J. *in review*. Surface and subsurface biogeochemical impacts of the 2019 Gulf of Alaska marine heatwave. *Geophysical Research Letters*.

In print:

5. **Mogen, S.C.**, Lovenduski, N.S., Yeager, S.G., Capotondi, A., Bograd, S.J., Hazen, E.L., Pozo Buil, M., Jacox, M.G., Di Lorenzo, E., Kim, W., Rosenbloom, N.R.. (2024) Multi-month forecasts of marine heatwaves and ocean acidification extremes. *Nature Geoscience*, 17, 1261–1267.
<https://doi.org/10.1038/s41561-024-01593-0>
4. **Mogen, S.C.**, Lovenduski, N.S., Yeager, S., Keppler, L., Sharp, J., Bograd, S.J., Quiros, N.C., Di Lorenzo, E., Hazen, E.L., Jacox, M.G., & Pozo Buil, M. (2023). Skillful Multi-Month Predictions of Ecosystem Stressors in the Surface and Subsurface Ocean. **Earth's Future**, 11(11).
<https://doi.org/10.1029/2023ef003605>

3. Dukes, E., Cheng, S., **Mogen, S.C.**, Galloway, J., Leach, A., Trimble, A. R., Pettit, A., Compton, J., & Pennino, M. (2023). Footprints in Action: How UVA Is Managing Its Sustainability Stewardship. *Sustainability and Climate Change*, 16(1), 48–63.
<https://doi.org/10.1089/scc.2022.0067>
2. **Mogen, S.C.**, Lovenduski, N.S., Dallmann, A.R., Gregor, L., Sutton, A.J., Bograd, S.J., Quiros, N. C., Di Lorenzo, E., Hazen, E.L., Jacox, M.G., Pozo Buil, M., & Yeager, S. (2022). Ocean Biogeochemical Signatures of the North Pacific Blob. *Geophysical Research Letters*, 49(9).
<https://doi.org/10.1029/2021gl096938>
1. Yeager, S.G., Rosenbloom, N., Glanville, A.A., Wu, X., Simpson, I., Li, H., Molina, M.J., Krumhardt, K., **Mogen, S.C.**, Lindsay, K., Lombardozzi, D., Wieder, W., Kim, W.M., Richter, J.H., Long, M., Danabasoglu, G., Bailey, D., Holland, M., Lovenduski, N.S., Strand, W., & King, T. (2022). The Seasonal-to-Multiyear Large Ensemble (SMYLE) prediction system using the Community Earth System Model version 2. *Geoscientific Model Development*, 15(16), 6451–6493.
<https://doi.org/10.5194/gmd-15-6451-2022>

HONORS AND AWARDS

Community Earth System Model Graduate Student Award	2024
Atmospheric and Oceanic Sciences Departmental Service Award	2020-2024
Earth and Space Science Poster Session, First in Oceanography	2021
NSF Graduate Research Fellowship, Honorable Mention	2021
Atmospheric and Oceanic Sciences Departmental Teaching Assistant Award	Fall 2020
Wallace-Poole Prize, University of Virginia	2020
Echols Scholar, University of Virginia	2016-2020
Ernest F. Hollings Scholar, NOAA	2018

SERVICE & OUTREACH

Peer-Reviewer for: <i>Geophysical Research Letters</i> , <i>Earth's Future</i> , <i>Climate Dynamics</i> , <i>Journal of Geophysical Research: Oceans</i> , <i>Progress in Oceanography</i>	
ATOC Departmental Colloquium Committee	2022-2024
Justice, Equity, Diversity and Inclusion Committee, INSTAAR	
Co-Lead, Committee for an Inclusive Graduate Student Body	2022-2023
Co-Lead and Co-Developer, Department Microaggression and Bias Seminars	2021-Present
Co-Lead and Organizer, ATOC REU Python Bootcamp	2021-2024
Justice, Equity, Diversity and Inclusion Committee, ATOC	2020-present
Committee Chair	2023-present
ATOC Graduate Program Application Mentor	2020-present

TEACHING

University of Colorado Boulder	
Lead Teacher, <i>ATOC REU Python Bootcamp</i>	Summer 2021-2024
Teaching Assistant, <i>ATOC 3070: Introduction to Oceanography</i>	Fall 2020, Spring 2023
Teaching Assistant, <i>ATOC 1060: Our Changing Environment</i>	Spring 2021
Guest Lecturer, <i>ATOC 5200: Biogeochemical Oceanography</i>	2023, 2024

MENTORSHIP

Nahir Guadalupe

2021

CU Boulder ATOC REU - *Small-scale variations in marine phytoplankton abundance in the California Current System*

Valeria Perez-Rivera

2022

CU Boulder Summer Multicultural Access to Research Training (SMART) Program - *Ocean Acidification Over the Past Four Decades*

WORKSHOPS, SYMPOSIA, SUMMER SCHOOLS, AND FIELD EXPERIENCES

7. Dissertation Symposium in Chemical Oceanography (DISCO), Lihue, HI* Oct. 2024
6. ICTP/CLIVAR Summer School on Marine Heatwaves. Trieste, Italy.* July, 2023
5. Surface Ocean - Lower Atmosphere Summer School 2023. Mindelo, Cape Verde.* June, 2023
4. U.S. GO-SHIP Occupation of Line A16N* Spring 2023
3. The 5th International Symposium on the Ocean in a High CO₂ World from U.S. Ocean Carbon and Biogeochemistry, Lima, Peru Sep. 2022
2. Columbia University Center for Learning the Earth with Artificial Intelligence and Physics (LEAP) Momentum Bootcamp on Climate Data Science. Briarcliff Manor, New York.* May 2022
1. National Center for Atmospheric Research (NCAR) Advanced Study Program (ASP) Colloquium 2021. The Science of S2S Predictions. Virtual.* July 2021, 2022

* indicates invited participant

SELECTED PRESENTATIONS

Invited Talks:

4. **Mogen, S.** Extremes and predictability of ocean biogeochemical ecosystem stressors. NOAA Physical Sciences Laboratory Seminar. February, 2025.
3. **Mogen, S.** Multi-month forecasts of marine heatwaves and ocean acidification extremes. CESM Workshop. June, 2024.
2. **Mogen, S.** Skillful Multi-month Predictions of Ecosystem Stressors in the Surface and Subsurface Ocean. BGC Argo Science Meeting. November, 2023.
1. **Mogen, S.** Near-term predictions of (sub)surface biogeochemical state. BGC Argo Science Meeting. September, 2022.

Conference Talks:

14. **Mogen, S.** Lovenduski, NS, Yeager, S, Capotondi, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Kim, W, Rosenbloom, NR. "Multi-month forecasts of marine heatwaves and ocean acidification extremes." OceanPredict Symposium 2024. Paris, France. November, 2024.
13. **Mogen, S.** Lovenduski, NS, Yeager, S, Capotondi, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Kim, W, Rosenbloom, NR. "Multi-month forecasts of marine heatwaves and ocean acidification extremes." Ocean Sciences Meeting. New Orleans, LA. February, 2024.
12. **Mogen, S.** Lovenduski, NS, Yeager, S. "Multi-month predictions of extremes in ocean temperature and acidity." ICTP/CLIVAR Summer School on Marine Heatwaves. Trieste, Italy. July, 2023.
11. **Mogen, S.** Lovenduski, NS, Yeager, S, Keppler, L., Sharp, J., Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Predicting marine stressors multiple months in advance." SOLAS Summer School. Mindelo, Cape Verde. June, 2023.

10. **Mogen, S**, Lovenduski, NS, Yeager, S, Keppler, L, Sharp, J, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Predicting marine stressors multiple months in advance . " NOAA/CPO/CVP Webinar Series - Decadal Variability and Predictability. Virtual. June, 2023.
9. **Mogen, S**, Lovenduski, NS, Yeager, S, Keppler, L, Sharp, J, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Skillful predictions of multiple marine stressors in the surface and subsurface ocean." CESM Ocean Model Working Group Meeting. Boulder, CO. February, 2023.
8. **Mogen, S**, Lovenduski, NS, Yeager, S, Keppler, L, Sharp, J, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Near-term predictions of (sub)surface marine stressors." 5th International Symposium on the Ocean in a High CO2 World. Lima, Peru. September, 2022.
7. **Mogen, S**, Lovenduski NS, Yeager, S. "Multi-Month Predictions of Ocean Acidification in CESM SMYLE." CESM Earth System Prediction Working Group Meeting. Boulder, CO. February, 2022.
6. **Mogen, S**, Lovenduski NS, Yeager, S, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Ocean biogeochemical signatures of the North Pacific Blob." ATOC Departmental Colloquium. Boulder, CO. February, 2022.
5. **Mogen, S**, Lovenduski NS, Yeager, S, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Ocean biogeochemical signatures of the North Pacific Blob." Ocean Sciences Meeting. Virtual. February, 2022.
4. **Mogen, S**, Lovenduski NS, Yeager, S, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Ocean biogeochemical signatures of the North Pacific Blob." WCRP Workshop on Extremes in Climate Prediction Ensembles (ExCPEns). Virtual. October, 2021.
3. **Mogen, S**, Lovenduski NS, Yeager, S, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Ocean biogeochemical signatures of the North Pacific Blob." Graduate Climate Conference. Virtual. October, 2021.
2. **Mogen, S**, Lovenduski NS, Yeager, S, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Ocean biogeochemical signatures of the North Pacific Blob." Eastern Pacific Ocean Conference. Fallen Leaf Lake, CA. September, 2021.
1. **Mogen, S**, Pilcher, D, Cross, JN. "Outlook on Changes in Bering Sea Oxygen Cycling." Hollings Scholar Symposium. Silver Spring, MD. August, 2019.

Conference Posters:

7. **Mogen, S**, Lovenduski NS, "Multi-month forecasts of marine heatwaves and ocean acidification extremes." AGU. Washington, D.C. December, 2024.
6. **Mogen, S**, Lovenduski NS, "Multi-month forecasts of marine heatwaves and ocean acidification extremes." Earth System & Space Science (ESSS) Poster Session: Boulder, CO. December, 2023.
5. **Mogen, S**, Lovenduski, NS, Yeager, S, Keppler, L., Sharp, J., Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Skillful predictions of multiple surface and subsurface marine stressors." SOLAS Summer School. Mindelo, Cape Verde. June, 2023.
4. **Mogen, S**, Lovenduski, NS, Yeager, S, Keppler, L., Sharp, J., Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Skillful predictions of multiple surface and subsurface marine stressors." American Meteorological Society Meeting. Denver, CO. January, 2023.

3. **Mogen, S**, Lovenduski, NS. "Predictions of multiple ocean stressors in CESM-SMYLE in Coral Reef Regions." US Clivar Workshop on Societally-Relevant Multi-Year Climate Predictions. Boulder, CO. March, 2022.
2. **Mogen, S**, Lovenduski NS, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Ocean biogeochemical signatures of the North Pacific Blob." Earth System Space Science (ESSS) Poster Session: Boulder, CO. December, 2021.
1. **Mogen, S**, Doney, S, Pilcher, D, Cross, JN. "Outlook on Changes in Bering Sea Oxygen Cycling." Ocean Sciences Meeting: San Diego, CA. February 2020.